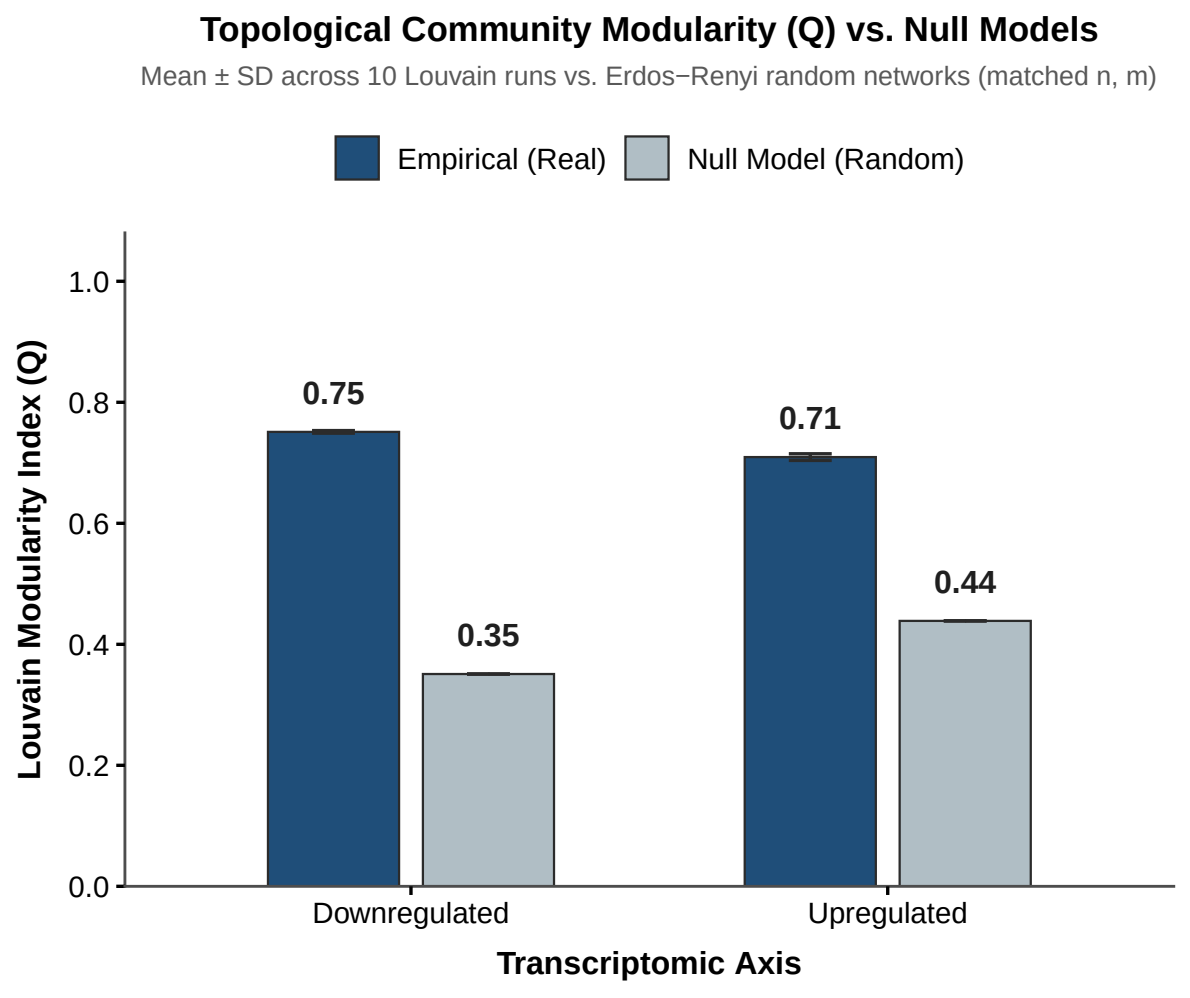
Topological Community Modularity (Q) vs. Null Models
Mean ± SD across 10 Louvain runs vs. Erdos−Renyi random networks (matched n, m)
Empirical (Real)
Null Model (Random)
Louvain Modularity Index (Q)
1.0
0.8
0.6
0.4
0.2
0.0
0.75
0.35
0.71
0.44
Downregulated
Upregulated
Transcriptomic Axis